Shannon Meets Random Oracles: On the Power of Collision Resistant Hashing in Black-Box Constructions

Anonymous Submission to Eurocrypt 2013

October 29, 2012

Abstract

Shannon in 1949 showed that any statistically secure encryption scheme between Alice and Bob for m-bit messages requires a shared private-key of length $k \geq m$. In this work we extend Shannon's theorem to the setting where Alice and Bob communicate interactively while they share an arbitrary long key R with random access (i.e. a random oracle), read at most k bits of R, and are allowed a small probability of error.

As applications, we obtain lower-bounds on the efficiency of black-box constructions of *private-key amplification* schemes from a variety of cryptographic primitives (including collision-resistant hash functions, commitment-schemes, trapdoor-permutations, etc). In a private-key amplification scheme Alice and Bob share a key of length k and want to obtain a (pseudorandom) key of length $m > k$ through interaction. As corollaries, we derive lower-bounds on the efficiency of black-box constructions of public-key or private-key encryption schemes as well as string oblivious transfers.

Keywords: Statistical Secrecy, Black-Box Constructions, Encryption, Key-Agreement.

1 Introduction

In 1949 Shannon [Sha49] showed that to achieve information theoretically secure encryption schemes the key should be as long as the message. This strong barrier to obtain nontrivial encryption has affected the main theme in foundations of cryptography to base the security of the protocols over *computational assumptions*. This approach naturally has led to well studied *cryptographic primitives* e.g., one-way functions, trapdoor permutations, oblivious transfer, etc, whose existence is assumed (as a computational assumption) to guarantee the security of the cryptographic task.

Black-box constructions and separations. Most of the cryptographic constructions are *fully black-box* in the sense that: the algorithms implementing the construction $\mathcal{Q}$ use some cryptographic primitive $\mathcal{P}$ (e.g., one-way functions) only as an oracle, and **(2)** the *security reduction* takes any adversary Adv who breaks $\mathcal{Q}^{\mathcal{P}}$ also as an oracle and produce an attack against $\mathcal{P}$ (see Section A.2 for formal definitions). In addition to capturing most of our techniques in cryptography, black-box constructions are considered important due to the (usual) inefficiency of non-black-box techniques. Impagliazzo and Rudich [IR89] were the first to show that black-box techniques in fact *cannot* construct public-key encryption based only on private-key encryption, thus "separating" the two worlds [Imp95]. This seminal work led to a sequence of result (e.g., [Sim98, GKM+00, BPR+08] to name a few) known as "black-box separations" that study the black-box power of cryptographic primitives. The work of Reingold, Trevisan, and Vadhan [RTV04] formalized black-box constructions further and established a taxonomy for the field.

Efficiency of black-box constructions. The works of Kim, Simon and Tetali [KST99] and Gennaro and Trevisan [GT00] initiated a closely related line of work studying how *efficient* black-box constructions can be (which is a meaningful question even if achieving a black-box construction is possible). Many subsequent works followed (e.g., [GGK03, HK05, BMG07, HHRS07] to name a few) among which the most relevant to us is the work of Gennaro, Gertner, and Katz [GGK03] that proved lower-bounds on the efficiency of basic cryptographic tasks, such as private-key and public-key encryption, using basic primitives such as one-way or trapdoor permutations (which are indeed sufficient to achieve private-key and public-key encryption in a black-box way).

Power of collision-resistant hashing. The set of primitives studied in the lower-bounds of [GGK03] do not include a basic primitive: collision resistant hash functions (CRHFs). However, CRHFs have been extremely useful to achieve efficient cryptographic primitives. For example, black-box constructions of two-message statistically hiding commitments can be obtained from CRHFs [DPP97] while by using one-way (or even trapdoor) permutations as a black-box an almost linear number of rounds is required [HHRS07]. To obtain efficient protocols, the ideal solution might use several primitives and thus it is possible that CRHFs *together with* others primitives lead to very efficient schemes. A main question we address in this work is the following:

> *How efficient could black-box constructions of private-key or public-key encryption be when we use collision resistant hash functions together with other primitives?*

More generally, an interesting question is to what extent these black-box separations extend to scenarios where natural "non-degeneracy" requirements are imposed on the underlying primitives—e.g., we may want to require that a one-way function "mixes" its input; the fact that collision-resistance is not covered by the previous lower-bounds is due to the fact that collision-resistant hash functions are inherently input mixing. We will discuss this issue shortly.

1.1 Our Results

We prove lower-bounds on the efficiency of a variety of cryptographic primitives based on basic cryptographic primitives while collision resistant hash functions are also available as black-box. We prove our lower-bounds in a unified way by employing a central primitive that we call *private-key amplification* schemes defined as follows.

Definition 1.1 (Private-Key Amplification)**.** A *k-to-m private-key amplification* scheme is a two-party protocol between poly(n)-time parties Alice and Bob in which they receive common inputs: a k-bit random string r and 1^n where n is the security parameter, and after the interaction Alice and Bob output X_A and X_B and the following two holds: **(1)** $\Pr[X_A = X_B] = 1 - \mathrm{negl}(n)$ and **(2)** the random variables (Y, X_A) and (Y, U_m) are computationally indistinguishable where Y is the transcript of the interaction and U_m is a uniform m bit string sampled *independently* of Y.

We prove the following lower-bound on the efficiency of private-key amplification schemes.

Theorem 1.2. *There is no fully black-box construction of r-round k-to-m private-key amplification schemes with $o(\frac{m-k}{\log n}) - r$ oracle queries to: collision-resistant hash functions, one-way permutations, trapdoor-permutations, semi-honest bit oblivious transfer, and commitment schemes.*

Since private-key and public-key encryption schemes can be interpreted as variants of private-key amplification schemes with only one or two rounds, and since string oblivious transfer implies interactive encryption schemes [GKM+00], we obtain the following corollary.

Corollary 1.3. *There is no fully black-box construction of semantically secure private-key encryption schemes with key length k and m bit messages using only $o(\frac{m-k}{\log n})$ oracle calls to CRHFs. For semantically secure public-key encryption schemes and oblivious transfer for m-bit messages there is no fully black-box construction using only $o(m/\log n)$ oracle calls to CRHFs.*

Corollary 1.3 extends to all the primitives mentioned in Theorem 1.2 as well.

1.2 Technique: Generalizing Shannon's Lower-Bound

The central tool in proving our lower-bounds is an extension of Shannon's information theoretic lower-bound on the key length of statistically secure encryption schemes. Our result generalizes Shannon's result in two ways: **(1)** the parties access a huge random string (i.e. a private random oracle) while they read only a few bits of it, and **(2)** instead of Alice sending a (random) message to Bob, they jointly sample messages x_A and x_B which are equal with high probability.

Theorem 1.4. *Suppose Alice and Bob interact while accessing a random oracle O and the total number of bits of O read by either of Alice or Bob is at most k. Suppose X_A (resp. X_B) denote Alice's (resp. Bob's) output at the end of the interaction and let Y denote the transcript of interaction. If $\Pr[X_A = X_B] \geq 1-\gamma$, then it holds that $\Delta((X_A, Y), (U_m, Y)) \geq 1 - 2^{k-m} - \gamma$ where U_m is uniformly distributed over $\{0,1\}^m$ independently of the transcript Y and $\Delta(\cdot,\cdot)$ denotes statistical distance.*

It is easy to see that if X_A is Alice's message in an encryption scheme (rather than her sampled output), the condition $\Delta((X_A, Y), (U_m, Y))$ can be turned into an actual attack against the statistical security of the encryption scheme (see Lemma A.3).

Private random oracle. The works of [IR89, BMM09] showed that any key agreement protocol in the random oracle model in which Alice and Bob ask k oracle queries can be broken by an eavesdropping algorithm Eve who asks $O(k^2)$ oracle queries. Our Theorem 1.4, however, is proved in a different model where the random oracle is *private* and is not accessed by the eavesdropper. However, for our attacker to succeed the parties Alice and Bob should ask fewer bits from their oracle than the key they agree on (otherwise there is a trivial secure protocol in which the parties read the same bits from the oracle and use them as key).

Using Theorem 1.4 to derive lower-bounds from collision-resistant hashing. Here we describe the high level outline of the proof of Theorem 1.2 using Theorem 1.4 when the primitive used by the key-amplification scheme is only CRHFs. (See Section 3 for a more detailed exposition of the proof for the general case).

At a very high level, we employ the general separation paradigm whose idea dates back to [IR89]. We find a randomized oracle O_{crh} which is collision-resistant against any poly(n)-query adversary, while any key-amplification protocol Π which asks only "a few" queries to O_{crh} can be broken by an adversary Adv who ask only poly(n) queries to O_{crh}. This way, the existence of fully black-box constructions implies that the adversary Adv can be turned into a poly(n)-query attack against the collision-resistance of O_{crh} which is impossible (see Lemma A.7). Our construction of O_{crh} is in fact very simple: Given any query x, output a random string of length slightly super-logarithmic $t(n) = \omega(\log n)$. The fact that $2^t(n) = n^{\omega(1)}$ guarantees that O_{crh} is indeed collision resistant (see Claim 3.2). The more crucial point is that O_{crh} is indeed a random oracle whose queries return "a few" random bits. Thus, if the total number of queries q asked by Alice or Bob (in proton Π) to O_{crh} is "small", then the total random bits of O_{crh} read by Alice or Bob will also be small. Finally, since we can think of the original k-bit private key key also as part of a random oracle $O = (\mathsf{key}, O_{\mathsf{crh}})$, we conclude that Alice and Bob are asking a few bits of the random oracle O. However, the number of bits read off O should be at least m (which constitutes a lower-bound on q), otherwise Theorem 1.4 would indeed imply the existence of Adv who breaks Π without asking any query to O.

To extend our Theorem 1.2 to one-way and trapdoor permutations we employ techniques from [GGK03] which we will describe and compare to our techniques in the following.

Comparison with techniques of [GGK03]. The lower-bounds of [GGK03] heavily rely on the fact that it is possible to have a highly degenerate oracle O that **(1)** O is a secure one-way (and trapdoor) permutation against any poly(n)-query attacker and **(2)** the answers of O essentially depends only on (almost) logarithmically many bits of the input. For example, for the case of one-way permutations, any query $x = (x_1, x_2)$ where $x_1 = \omega(\log n)$ is mapped to $O(x) = (y, x_2)$ where y is obtained from x_2 using a random permutation. This way, by only having the "short" string (x_1, y) one can conclude the answer of O to any other query that starts with x_1. [GGK03] showed the use of any such oracle O in an interactive protocol can be eliminated by using a not-too-big private key shared between the parties.[1] We emphasize that the oracles of this form are not "input mixing" in the sense that they split their input into two parts (x_1, x_2) where x_1 is short and is the the only part that matters and is needed to encode oracle query-answers.

Collision resistant hash functions, on the other hand, are highly non-degenerate and do not fall into the category above. In particular, suppose an oracle O implements a compressing hash function $h\colon \{0,1\}^n \mapsto \{0,1\}^{n/2}$ while only $\omega(\log n)$ bits of the input are randomly mapped to $\omega(\log n)$ bits of the output, and the remaining $n - \omega(\log n)$ input bits are deterministically mapped to the remaining $n/2 - \omega(\log n)$ bits of the output. In this case a (computationally unbounded) algorithm that

[1] see Section 3 for an exposition of how [GGK03] does so and why doing so implies lower-bounds.

does not query O at all can find collisions in O by finding collisions in the deterministic part of h. Therefore, it seems one can not apply the techniques of [GGK03] to eliminate a poly(n)-query secure oracle O from a protocol. Not surprisingly, our oracle O_{crh} implementing CRHFs described above is highly "input mixing" in the sense that all of the input bits randomly affect the output bits. Thus, our result also directly extend to show black-box separations from highly input-mixing one-way functions. (We leave it as an open question to determine whether similar statement can be showed also for input-mixing trapdoor permutations.)

Notation. We usually use calligraphic letters for sets and capital letters for random variables, and same lower-case letters for samples from the random variables (e.g., $y \xleftarrow{\$} Y$). By $\Delta(X, Y)$ we denote the statistical distance between the random variables X and Y. We write $X \equiv X'$ if X and X' have the same marginal distributions. For jointly distributed random variables, by $(X \mid Y = y)$ or more simply $(X \mid y)$ we refer to the distribution of X conditioned on $Y = y$.

2 Extending Shannon's Lower-Bound to Random Oracles

In this section we prove Theorem 1.4.

Theorem 1.4 (Restated)**.** *Suppose Alice and Bob interact while accessing a random oracle O and the total number of bits of O read by either of Alice or Bob is at most k. Suppose X_A (resp. X_B) denote Alice's (resp. Bob's) output at the end of the interaction and let Y denote the transcript of interaction. If $\Pr[X_A = X_B] \geq 1 - \gamma$, then it holds that $\Delta((X_A, Y), (U_m, Y)) \geq 1 - 2^{k-m} - \gamma$ where U_m is uniformly distributed over $\{0,1\}^m$ independently of the transcript Y and $\Delta(\cdot, \cdot)$ denotes statistical distance.*

Theorem 1.4 can be interpreted as a lower-bound on the key-length of statistically secure 0-to-m-bit key-agreement as defined in Definition 1.1 (which for simplicity we will simply call as m-bit key-agreements).

The non-interactive case. Note that a *non-interactive* m-bit key-agreement protocol is in fact a non-interactive encryption scheme in which x_A is Alice's message and the only message sent from Alice to Bob is the encryption of x_A. In Section B we prove lower-bounds on the query-complexity of *both* the sender and the receiver (which are both lower bounds on the key length k) for statistically secure non-interactive encryption schemes. This would allow us to lower-bound the efficiency of *both* parties in a black-box construction of private-key encryption scheme based on CRHFs, trapdoor permutations, etc.

We first give a proof for the special case of perfect completeness (i.e. $\gamma = 0$), and then will extend it to the general case where $\gamma > 0$.

2.1 Special Case of Perfect Completeness

In the following let V_A denote the view of Alice describing her private randomness as well as the oracle query-answer pairs observed by her during her execution, and define V_B similarly for Bob.

A special property that we will use in case of perfect completeness is that for every transcript of the interaction y and every fixed oracle O, there is at most one consistent output message x (for Alice and Bob). This is because if there are two pairs of (V_A, V_B) and (V'_A, V'_B) of views of Alice and Bob with output pairs (x_A, x_B) and (x'_A, x'_B), by perfect completes it holds that: $x_A = x_B$ and $x'_A = x'_B$. Moreover, by one more application of perfect completeness, and relying on the fact that V_A and V'_B are also consistent with (y, O), it holds that $x_A = x'_B$, and thus $x_A = x_B = x'_A = x'_B$.

Claim 2.1. *For every transcript y it holds that $\Delta((X_A \mid Y = y), U_m) \geq 1 - 2^{k-m}$.*

Before proving Claim 2.1 we note that together with Lemma 2.2 below (whose proof can be verified by straightforward calculation) they imply Theorem 1.4 for special case of perfect completeness.

Lemma 2.2. *Let X_1, X_2, Y be three jointly distributed random variables. Then it holds that $\Delta((X_1, Y), (X_2, Y)) = \mathbb{E}_{y \leftarrow Y} \Delta((X_1 \mid y), (X_2 \mid y))$.*

By Lemma 2.2 we have $\Delta((X_A, Y), (U_m, Y)) = \mathbb{E}_{y \leftarrow Y} \Delta((X_A \mid Y = y), U_m)$ because U_m is sampled uniformly and independently of the encryption Y. Claim 2.1 shows that for *every* y it holds that $\Delta((X_A \mid y), U_m) \geq 1 - 2^{k-m}$ which implies that $\Delta((X_A, Y), (U_m, Y)) \geq 1 - 2^{k-m}$ as well.

Proof of Claim 2.1. We claim that for every encryption y the set of consistent Alice's outputs has size at most 2^k. This shows that the distribution of X_A conditioned on the encryption y has statistical distance at least $1 - 2^{k-m}$ from the uniform distribution over $\{0,1\}^m$.

Recall that perfect completeness implies that for every oracle O, the pair (O, y) determines at most one consistent message x_A. If the support of the random variable $(X_A \mid y)$ has size more than 2^k, then for at least one message x_A, the total number of oracles consistent with (x, y) is at least one and at most $2^{n-k} - 1$ where n is the length of the oracle. But, this is not possible, because for every oracle O that is consistent with (V_A, V_B) one can obtain at least 2^{n-k} different oracles that are consistent with the same (V_A, V_B) by fixing the bits of O that are read by Alice and Bob and changing any other of $n - k$ bits arbitrarily.

□

2.2 The General Case

In this section we prove Theorem 1.4 in its general form that allows imperfect completeness $1 - \gamma$. Similarly to the case of perfect completeness, we prove a stronger statement by showing that $\Delta((X_A \mid Y = y), U_m) \geq 1 - 2^{k-m} - \gamma$ for every transcript y.

For every y, let $\mathsf{com}(y) = \Pr[X_A = X_B \mid y]$ be the completeness of the protocol conditioned on y. The completeness of the protocol $1 - \gamma$ is indeed equal to $\mathbb{E}_{y \leftarrow Y}[\mathsf{com}(y)]$, and so the following claim implies that $\Delta((X_A \mid Y = y), (U_m, Y)) \geq 1 - 2^{k-m} - \gamma$ (which in turn proves Theorem 1.4) by taking average over $y \xleftarrow{\$} Y$.

Claim 2.3. *For every transcript $y \xleftarrow{\$} Y$ it holds that $\Delta((X \mid Y = y), U_m) \geq \mathsf{com}(Y) - 2^{k-m}$.*

In the following we prove Claim 2.3.

Notation. Let $\varepsilon = \Delta((X_A \mid Y = y), U_m), M = 2^m, K = 2^k$ and $N = 2^n$ where n is the length of the oracle. Therefore our goal of proving Claim 2.3 is equivalent to showing that $\mathsf{com}(Y) \leq K/M + \varepsilon$. By $\mathbb{E}_{P|q}[\cdot]$ we denote the expectation where the random variable P is chosen from its distribution conditioned on $Q = q$ (for some correlated random variable Q). For any fixed transcript y and oracle O, we let $\mathsf{val}_y(O) = \max_x \Pr[X_A = x \mid y, O]$.

Claim 2.4. *It holds that $\mathsf{com}(y) \leq \mathbb{E}_{O|y} \mathsf{val}_y(O)$.*

Proof. In the following we might simply write $\mathsf{val}(O)$ when y is fixed and clear from the context.

We claim that $\Pr[X_A = X_B \mid O, y] \leq \mathsf{val}(O)$. That holds because conditioned on any fixed (O, y) V_A and V_B (and consequently X_A and X_B) are independent random variables, and two independent

random variables X_A, X_B would be equal at most with probability $\max_x \Pr[X_A = x]$. By taking average over the choice of O it holds that:

$$\mathsf{com}(y) = \Pr[X_A = X_B \mid y] = \mathop{\mathbb{E}}_{O|y} \Pr[X_A = X_B \mid O, y] \leq \mathop{\mathbb{E}}_{O|y} \mathsf{val}(O).$$

□

Claim 2.4 and the following claim together imply Claim 2.3.

Claim 2.5. *For every transcript y it holds that $\mathbb{E}_{O|y} \mathsf{val}(O) \leq K/M + \varepsilon$.*

In the following we will prove Claim 2.5 using a combinatorial lemma. For a bipartite graph $G = ((U, V), E)$ with vertex nodes (U, V) and edges E, for any vertex $u \in U \cup V$ by $\mathrm{adj}(u)$ we denote the set nodes that are adjacent to u, and by $\deg(u)$ we denote the degree of u. Looking ahead, we use a graph G such that the nodes in U encode the description of some oracle and the nodes in V denote a pair of views for Alice and Bob.

Lemma 2.6 (Combinatorial Lemma)**.** *Let $G = ((U, V), E)$ be a bipartite graph in which every node $v \in V$ has a color $\mathsf{col}(v) \in [M]$, and let $\mathsf{val}(u) = \max_{c \in [M]} \Pr_{v \xleftarrow{\$} \mathrm{adj}(u)}[\mathsf{col}(v) = c]$. Suppose we have:*

1. *$(\mathsf{col}(v) \mid (u, v) \xleftarrow{\$} E)$ (i.e. the distribution of the color of v where (u, v) is a random edge) is ε-close to the uniform distribution over $[M]$, and*
2. *$\deg(v) \geq |U|/K$ for every $v \in V$.*

Then it holds that $\mathbb{E}_{(u,v) \xleftarrow{\$} E} \mathsf{val}(u) \leq K/M + \varepsilon$.

We first prove Claim 2.5 using Lemma 2.6.

Proof of Claim 2.5. For the fixed encryption y define the bipartite graph $G = ((U, V), E)$ as follows. The nodes in U consist of all possible oracles O (conditioned on y), and every node in V is described by a pair of views (V_A, V_B) for Alice and Bob that are consistent with y. We connect $O \in U$ to $(V_A, V_B) \in V$ iff (V_A, V_B) is consistent with O (i.e. all of (V_A, V_B, y, O) are consistent together). Finally let $\mathsf{col}(v) = x_A \in [M]$ where x_A is the output of Alice described in $v = (V_A, V_B)$.

W.l.o.g we assume that Alice and Bob use random seeds of the same length ℓ. In this case for every transcript y it holds that $\Pr[V_A, V_B, O \mid y] = \Pr[V_A, V_B, O, y]/\Pr[y] = 2^{-2\ell} \cdot 2^{-n}/\Pr[y]$ which is independent of the nodes (V_A, V_B) and O. Therefore, the distribution $(V_A, V_B, O \mid y)$ can be sampled by sampling an edge $e \xleftarrow{\$} E$ in G *uniformly* at random. This shows that for $u \in U$ corresponding to the oracle O it indeed holds that the function $\mathsf{val}(\cdot)$ for oracles and the nodes in U are defined equally: $\mathsf{val}(O) = \mathsf{val}(u)$, and also the first condition of Lemma 2.6 holds.

For any pair of views (V_A, V_B) that are consistent with O and in which Alice and Bob together read at most k bits of O, the pair of views (V_A, V_B) will be consistent with at least $2^{n-k} = N/K$ many more oracles (different from O) by fixing O over the read bits and changing it elsewhere. In addition, the total number of the oracles is 2^n implying that $|U| \leq 2^n = N$, and therefore the second condition of Lemma 2.6 holds as well. Having both conditions satisfied, Claim 2.5 follows from Lemma 2.6 directly. □

Finally we will prove our combinatorial Lemma 2.6.

Proof of Lemma 2.6. We first prove Lemma 2.6 for a special case where $\Delta((\mathsf{col}(v) \mid (u,v) \xleftarrow{\$} E), U_m) = 0$ (i.e. $\varepsilon = 0$), and then we will then prove Lemma 2.6 in its full fledged form, by reducing the problem to the case of $\varepsilon = 0$. Note, however, that the special case already assumes that $|V|/K$ is an integer (otherwise the number of V nodes with different colors cannot be equal); thus the extension to the general case is not quite straightforward and as we will see needs extending the problem to fractional colors.

Special case of $\varepsilon = 0$. First note that the degree on every node in V is at least $|U|/K$, so $|E| \geq |V| \cdot |U|/K$. In addition, by $\Delta((\mathsf{col}(v) \mid (u,v) \xleftarrow{\$} E), U_m) = 0$, for every $u \in U$, it holds that $\max_{c\in[M]} |\{\mathsf{col}(v) = c \wedge v \in \mathrm{adj}(u)\}| \leq \max_{c\in[M]} |\{\mathsf{col}(v) = c\}| = |V|/M$. Therefore:

$$\begin{aligned}\mathop{\mathbb{E}}_{(u,v)\xleftarrow{\$}E} \mathsf{val}(u) &= \sum_{u\in U} \frac{\deg(u)}{|E|} \cdot \frac{\max_{c\in[M]} |\{\mathsf{col}(v) = c \wedge v \in \mathrm{adj}(u)\}|}{\deg(u)} \\ &= \sum_{u\in U} \frac{\max_{c\in[M]} |\{\mathsf{col}(v) = c \wedge v \in \mathrm{adj}(u)\}|}{|E|} \leq \sum_{u\in U} \frac{|V|/M}{|V| \cdot |U|/K} = K/M.\end{aligned}$$

General case. To handle the general case where $\varepsilon > 0$ (and $|V|/M$ is not necessarily an integer), we cannot anymore achieve a uniformly distributed coloring by sampling the color of a random vertex in V. A natural idea is to use *fractional colors*. Namely, we allow every node to carry a fraction of every color while the summation of these fractions still summon to one unit of color. Following this idea, we will prove a generalization of Lemma 2.6 as follows. We allow the color of a vertex $v \in V$ to be a *distribution* over $[M]$ represented by a vector $\mathsf{col}(v) = (\mathsf{col}(v)_1, \cdots, \mathsf{col}(v)_M)$ with nonnegative components such that $\sum_{i\in[M]} \mathsf{col}(v)_i = 1$. By $i \xleftarrow{\$} \mathsf{col}(v)$ we denote a random selection of a color in $[M]$ that chooses i with probability $\mathsf{col}(v)_i$. In this generalization we define $\mathsf{val}(u) = \frac{1}{\deg(u)} \cdot \max_{i\in[M]} \sum_{v\in\mathrm{adj}(u)} \mathsf{col}(v)_i$. The first condition of Lemma 2.6 shall be restated as $\Delta((c \mid c \xleftarrow{\$} \mathsf{col}(v), (u,v) \xleftarrow{\$} E), U_m) \leq \varepsilon$.

Consider the following way of changing a coloring $\mathsf{col} = \{\mathsf{col}(v) \mid v \in V\}$ into another coloring $\mathsf{col}' = \{\mathsf{col}'(v) \mid v \in V\}$: In each step we are allowed to pick a vertex v with color $\mathsf{col}(v) = (c_1^b, \cdots, c_M^b)$, choose two components c_i^b, c_j^b such that $c_i^b \geq \delta$, decrease c_i^b by δ, and increase c_j^b by δ. We call such change of colors a *basic* change of weight δ over the vertex v.

Claim 2.7. *If $\Delta((i \mid i \xleftarrow{\$} \mathsf{col}(v), (u,v) \xleftarrow{\$} E), U_m) = \varepsilon$, then there is a sequence of basic changes with weights $\delta_1, \ldots, \delta_t$ over vertices $v_1, \ldots, v_t$ that transforms col into another coloring col' such that:*

1. $(i \mid i \xleftarrow{\$} \mathsf{col}'(v), (u,v) \xleftarrow{\$} E) \equiv U_m$.
2. $\varepsilon = \sum_{i\in[t]} \delta_i \deg(v_i)/|E|$.

To prove Claim 2.7 we use an extension of Lemma 9.4 in [Hol07]. In [Hol07] Holenstein showed that for any joint random variables (X, Y) if $\Delta(Y, Z)$ is "small", then there is a way to sample X and Z in a correlated way (denoted by (X, Z)) such that $\Delta((X, Y), (X, Z))$ remains "small" as well. Here we need a constructive version of this lemma. (See Appendix C for a proof.)

Lemma 2.8. *Let (X, Y) be a joint distribution over $[N] \times [M]$, and Z be a distribution over $[M]$. There is a sequence of random variables $Y \equiv Y_0, Y_1, \ldots, Y_t \equiv Z$ all correlated with X such that:*

- *For every $i \in [t]$ there are x_i, y_i, y_i' and δ_i such that:*
  1. $\Pr[(X, Y_{i-1}) = (x_i, y_i)] = \Pr[(X, Y_i) = (x_i, y_i)] - \delta_i$.

2. $\Pr[(X, Y_{i-1}) = (x_i, y_i')] = \Pr[(X, Y_i) = (x_i, y_i')] + \delta_i$.
3. $\Pr[(X, Y_{i-1}) = (x, y)] = \Pr[(X, Y_i) = (x_i, y_i)]$ *for all* $(x, y) \notin \{(x_i, y_i), (x_i, y_i')\}$.

- $\sum_i \Pr[X = x_i] \cdot \delta_i = \Delta(Y, Z)$.

Now we show how to conclude Claim 2.7 using Lemma 2.8.

Proof of Claim 2.7 using Lemma 2.8. This claim directly follows from Lemma 2.8 as follows. We let the random variable X (of Lemma 2.8) to be the distribution $(v \mid (u, v) \leftarrow E)$. For a given value of $X = v$, let $Y \mid X$ be the distribution $c \xleftarrow{\$} \mathsf{col}(v)$. We just have to note that $\Pr_{(u,v) \xleftarrow{\$} E}[v = v_0] = \deg(v_0)/|E|$ in which case Claim 2.7 follows from Lemma 2.8. □

Claim 2.9. *Any basic change of weight* δ *over* $v_0 \in V$ *changes* $\mathbb{E}_{(u,v) \xleftarrow{\$} E} \mathsf{val}(u)$ *at most by* $\frac{\delta \cdot \deg(v_0)}{|E|}$.

Proof. For simplicity first consider a basic change of weight 1 over a node $v_0 \in V$ (i.e. changing the color of v_0 completely) and let see how it affects $\mathbb{E}_{(u,v) \xleftarrow{\$} E} \mathsf{val}(u)$. The total number of edges (u, v) for which the color of v_0 might affect $\mathsf{val}(u)$ is $\sum_{u \in \mathrm{adj}(v_0)} \deg(u)$, and for each $u \in \mathrm{adj}(v_0)$ this effect is at most $1/\deg(u)$. Therefor, the total effect this basic change can make is at most $\deg(v_0)/|E|$. When a basic change is of weight δ, for every relevant node a, the effect will be at most $\delta/\deg(u)$, which results in the total effect $\delta \cdot \deg(v_0)/|E|$. □

Claims 2.7 and 2.9 together show that that if $\Delta((i \mid i \leftarrow \mathsf{col}(v), (u, v) \xleftarrow{\$} E), U_m) = \varepsilon$, we can always change the coloring col through a finite sequence of basic changes that change $\mathbb{E}_{(u,v) \xleftarrow{\$} E} \mathsf{val}(u)$ by at most ε while we reach a "uniform" coloring satisfying: $\Delta((i \mid i \leftarrow \mathsf{col}(v), (u, v) \xleftarrow{\$} E), U_m) = 0$. Thus, it is sufficient to show that whenever we get $\Delta((i \mid i \leftarrow \mathsf{col}(v), (u, v) \xleftarrow{\$} E), U_m) = 0$, it holds that $\mathbb{E}_{(u,v) \xleftarrow{\$} E} \mathsf{val}(u) \leq K/M$, which is what we prove in the following.

Since the degree on every node in V is at least $|U|/K$ we conclude that $|E| \geq |V| \cdot |U|/K$. In addition (since we are assuming assuming $\Delta((i \mid i \leftarrow \mathsf{col}(v), (u, v) \xleftarrow{\$} E), U_m) = 0$) for every $u \in U$, it holds that $\max_{i \in [M]} \sum_{v \in \mathrm{adj}(u)} \mathsf{col}(v)_i \leq \max_{i \in [M]} \sum_{v \in V} \mathsf{col}(v)_i = |V|/M$. Therefore it holds that

$$\begin{aligned}
\mathop{\mathbb{E}}_{(u,v) \xleftarrow{\$} E} \mathsf{val}(u) &= \sum_{u \in U} \frac{\deg(u)}{|E|} \cdot \frac{\max_{i \in [M]} \sum_{v \in \mathrm{adj}(u)} \mathsf{col}(v)_i}{\deg(u)} \\
&= \sum_{u \in U} \frac{\max_{i \in [M]} \sum_{v \in \mathrm{adj}(u)} \mathsf{col}(v)_i}{|E|} \leq \sum_{u \in U} \frac{|V|/M}{|V| \cdot |U|/K} = K/M.
\end{aligned}$$

□

3 Lower-Bounds for Black-Box Private-Key Amplification

In this section we prove Theorem 1.2. More formally we prove the following:

Theorem 3.1. *There exists no fully black-box construction* Π *of* r*-round* k*-to-*m *private-key amplification schemes with completeness* $9/10$ *and* $q = o(\frac{m-k}{\log n}) - r$ *oracle queries to the following primitives: collision-resistant hash functions, statistically-hiding bit commitments, one-way permutations, trapdoor permutations, perfectly binding bit commitments, or semi-honest bit oblivious transfer.*

The CRHFs used in Theorem 3.1 could be even assumed by the construction to be optimally shrinking: $h \colon \{0,1\}^* \mapsto \{0,1\}^{\omega(\log n)}$. Note that if the image of the CRHF has only $O(\log n)$ bits it can be broken in polynomial time.

Reducing the primitives. To prove Theorem 3.1, w.l.o.g we can assume that the construction Π does not use (perfectly binding or statistically hiding) bit commitment schemes or semi-honest bit oblivious transfer directly. That is because these primitives can be obtained using other primitives listed in Theorem 3.1 in a black-box way and using a *constant* number of oracle as follows:

- Statistically hiding commitments using collision resistant hash functions [DPP97].
- Perfectly binding commitments using one-way permutations [GL89].
- Semi-honest bit oblivious transfer using trapdoor permutations [GMW87, Hai04].

Outline of the proof of Theorem 3.1. The big picture of the proof is as follows:

1. We start by assuming, for sake of contradiction, that there is a construction Π for which $q + r = o(\frac{m-k}{\log n})$.

2. We use two randomized oracles $(O_{\mathsf{tdp}}, O_{\mathsf{crh}})$ where O_{crh} is a randomized implementation of a family of collision-resistant hash functions and O_{tdp} is a randomized implementation of trapdoor permutations as well as one-way permutations. These oracles have the property that they use almost t number of random bits to answer any query for a slightly super logarithmic $t = \omega(\log n)$, yet no $\mathrm{poly}(n)$-query attacker can break their security with $1/\mathrm{poly}(n)$ probability. Let q_1 be the number of queries of the construction to O_{crh} and q_2 be the number of oracle queries to O_{tdp}.

3. We show that if $q_1 + q_2 + r = o(\frac{m-k}{\log n})$ then there is an attacker Adv who does not query the oracles $(O_{\mathsf{tdp}}, O_{\mathsf{crh}})$ and breaks the security of Π with advantage $1/3 > 1/\mathrm{poly}(n)$. After doing so, Theorem 3.1 immediately follows from our Lemma A.7 by using $\mathbf{O} = (O_{\mathsf{tdp}}, O_{\mathsf{crh}})$, $\mathcal{Q} =$ k-to-m private-key amplification, and $\mathcal{P} =$ a primitive which simultaneously provides all of the primitives CRHFs, TDPs, and OWPs. This step itself is done as follows.

   (a) Using the elegant techniques introduced in [GGK03] we get rid of the oracle O_{tdp} using a secret key key_2 of length $(q_2 + r) \cdot t$ shared between Alice and Bob (in addition to the k-bit key key_1 originally shared between them to start the protocol Π). This is done by Alice and Bob simulating the oracle O_{tdp} "in their head" and sending the queries and answers that they simulate to the other party for sake of consistency while these values are masked using a one-time pad encrypting with the help of key_2. We emphasize that we cannot use the oracle-elimination technique of [GGK03] to eliminate the CRHF oracle O_{crh} (see Section 1.2 for more discussions on this issue).

   (b) We can think of the keys $(\mathsf{key}_1, \mathsf{key}_2)$ as a (rather short) private random oracle O_{key} of length $\approx k + (r + q_2) \cdot t$ shared between Alice and Bob. Since the answers of the oracle O_{crh} each use $\approx t$ random bits, we conclude that Alice and Bob together are reading $\overline{k} \approx k + (q_1 + q_2 + r) \cdot t$ bits from their secret random oracle $(O_{\mathsf{key}}, O_{\mathsf{crh}})$.

   (c) Since we are assuming that $q_1 + q_2 + r = o(\frac{m-k}{\log n})$, and since $\overline{k} \approx k + (q_1 + q_2 + r) \cdot t$ by taking t sufficiently small we conclude that $\overline{k} < m$. By applying our Theorem 1.4 we get the desired attack Adv.

3.1 Proof of Theorem 3.1

Suppose there is a construction Π for which $q_1 + q_2 + r = \alpha(n) \cdot (\frac{m-k}{\log n})$ and $\alpha(n) = o(1)$ where q_1 is the number of queries to CRHFs and q_2 is the number of queries to OWPs and TDPs. Also let $\ell(n)$ be the length of the images of the CRHFs h (used by the construction Π) for security parameter n: $h\colon \{0,1\}^{\mathrm{poly}(n)} \times \{0,1\}^* \mapsto \{0,1\}^{\ell(n)}$ (the first input is the index of the hash function). Since

hash functions with logarithmic output length can be broken efficiently, w.l.o.g we assume that $\ell(n) = \omega(\log n)$.

Let $t(n) = \min(\frac{\log(n)}{10\alpha(n)}, \ell(n))$. Note that we have $t(n) = \omega(\log n)$. For brevity, when clear from the context we might denote $\alpha(n), \ell(n)$ and $t(n)$ simply by α, ℓ and t.

Consider the following two randomized oracles.

- O_{crh}: this oracle implements CRHFs as needed by Π as follows. Given any new query (d, x) (where d is the index of the hash function) choose t random bits y and output $(y, 0^{\ell-t})$ which is a string of length ℓ.

- O_{tdp}: this oracle implements trapdoor permutations using only t random bits for every query. This algorithm is due [GGK03]. To define O_{tdp}, let $o_{\mathsf{tdp}} = (g, f, f^{-1})$ be a random trapdoor permutation oracle with key, input, and output lengths all equal to t. Namely, $g(\cdot)$ is a random permutation over $\{0,1\}^t$, $f(z, \cdot)$ is a random permutation over $\{0,1\}^t$ for every z, and $f^{-1}(td, \cdot)$ is the inverse of the permutation $f(g(td), \cdot)$. The oracle O_{tdp} also consists of three sub oracles (G, F, F^{-1}) computed as follows (where $m > t$).

  - For input x of length $|x| = m$, let $x = (x_1, x_2)$ where $|x_1| = t$. Take $G(x) = (g(x_1), x_2)$.
  - For input (z, x) of lengths $|x| = |z| = m$, let $x = (x_1, x_2)$ and $z = (z_1, z_2)$ where $|x_1| = |z_1| = t$. Take $F(x) = (f(z_1, x_1), x_2)$.
  - For input (td, y) of lengths $|td| = |y| = m$, let $td = (td_1, td_2)$ and $y = (y_1, y_2)$ where $|td_1| = |y_1| = t$. Take $F^{-1}(td, y) = (f(td_1, y_1), y_2)$.

  It is easy to see that and $F^{-1}(td, \cdot)$ is the inverse of the permutation $F(G(td), \cdot)$.

Note that the oracles O_{crh} and O_{tdp} are defined for every security parameter n separately (in particular the same $t(n)$ bits are used as the relevant prefix for all input parameters m), and there might not be a single way to combine these oracles for all n. However, since our goal is to rule out fully-black-box constructions as opposed to relativizing reductions [RTV04], that is sufficient for us (based on Lemma A.7).

The following claim can be proved by lazy evaluation of the oracles O_{crh} and O_{tdp}.

Claim 3.2. *Any computationally unbounded* $\mathrm{poly}(n)$*-query attacker* A *can find collision in* O_{crh}*, break* $O_{\mathsf{tdp}} = (G, F, F^{-1})$ *as a trapdoor permutation, or break* G *as a one-way permutation only with* $\mathrm{negl}(n)$ *probability.*

Proof Sketch. Note that the randomized oracles O_{crh} and O_{tdp} are chosen at random after the algorithm A is fixed. We first show that A cannot find collisions in O_{crh} with $1/\mathrm{poly}(n)$ probability. The reason is that A has to find two inputs whose outputs match in their first t bits, but for any pair of queries collision happens only with probability $2^{-t} = \mathrm{negl}(n)$, and so by a union bound A cannot find collisions in O_{crh} with more than a negligible probability.

Now suppose A wants to invert G on a random image y of length n. This is equivalent to inverting a random y_1 over g where $|y_1| = t$. The claim easily follows due to randomness of the permutation g and the limited number of queries of A.

A similar arguments can also be made to show that breaking O_{tdp} as a trapdoor permutation requires breaking o_{tdp} which is a random trapdoor permutation. The latter can be shown to be impossible using similar standard arguments due to $2^t = n^{\omega(1)}$ and the fact that A only asks polynomially many queries. □

Claim 3.3. *If we use the oracles* $(O_{\mathsf{crh}}, O_{\mathsf{tdp}})$ *to implement the CRHFs, TDPs, and OWPs as required by the black-box construction* Π*, the security of* $\Pi^{(O_{\mathsf{crh}}, O_{\mathsf{tdp}})}$ *can be broken (as a* k*-to-*m *private-key amplification scheme) by a computationally unbounded adversary* Adv *who* does not *query the oracles* $(O_{\mathsf{crh}}, O_{\mathsf{tdp}})$ *at all.*

Concluding Theorem 3.1. Before proving Claim 3.3 we show how to use it together with Lemma A.7 to prove Theorem 3.1. In Lemma A.7 we let $\mathbf{O} = (O_{\mathsf{tdp}}, O_{\mathsf{crh}})$, let $\mathcal{Q}$ be k-to-m private-key amplification schemes, and let $\mathcal{P}$ be a primitive which provides all of the primitives CRHFs, TDPs, and OWPs. Namely, breaking $\mathcal{P}$ means that the adversary can break the security of at least one of the primitives CRHFs, TDPs, or OWPs implemented by $\mathcal{P}$. Firstly, Claim 3.2 shows that the randomized oracle $\mathbf{O} = (O_{\mathsf{tdp}}, O_{\mathsf{crh}})$ is secure against poly(n)-query attackers. Also the adversary Adv of Claim 3.3 breaks the security of Π with *zero* oracle queries (which is of course smaller than poly(n) queries). Finally, note that all of CRHFs, TDPs, and OWPs have security threshold zero (see Definition A.5). Therefore, Lemma A.7 proves that the black-box construction Π could not exist.

In the following we will focus on proving Claim 3.3.

Case of $q_2 = 0$. First we prove Claim 3.3 for the special case where Π only uses CRHFs (and thus the oracle O_{tdp} is not used by Π). The shared k-bit key given to Alice and Bob at the beginning of Π can itself be considered as a secret random oracle O_{key} which together with O_{crh} forms a larger random oracle $O = (O_{\mathsf{key}}, O_{\mathsf{crh}})$. The total number of bits read from O by either of Alice or Bob is $k + q \cdot t < k + q \cdot (\log n)/\alpha$. Since we are assuming $q < q + r \leq \alpha \cdot (\frac{m-k}{\log n})$, we conclude that the number of bits read from O is less than m. Therefore we can directly apply our attacker of Theorem 1.4 to obtain the attacker Adv needed for Claim 3.3 (who can break Π with advantage $1/3 > 1/2 - 1/10$).

General case of $q_2 > 0$. When $q_2 > 0$, we use techniques from [GGK03] to get rid of the oracle O_{tdp} at the cost of a longer private key between Alice and Bob while the new protocol $\overline{\Pi}$ is as secure as the original protocol Π.

First note that the oracle O_{tdp} is directly defined based on the oracle o_{tdp} which is defined for smaller input lengths (t and $2t$). Therefore, all we have to do is to find a way that Alice and Bob can simulate the oracle o_{tdp} in their head *consistently* without compromising the security.

Since o_{tdp} is a trapdoor permutation oracle, there is a simple algorithm for "lazy evaluation" of o_{tdp} due to [GGKT05]. Here we only describe the properties of this sampling procedure in an abstract way.

Lemma 3.4 (Described in the Proof of Theorem 5 in [GGKT05])**.** *There is an efficient algorithm* Eval *to simulate* o_{tdp} *as follows.*

- Eval *maintains a list* L *of query-answers of* o_{tdp} *sampled so far.*
- *On a new query* x*, either* Eval *concludes the answer to* x *based on* L*, or it tosses* t *new coins, obtains the answer, and adds it to* L*.*

Now consider the following protocol $\overline{\Pi}$ in which parties emulate the protocol Π while they simulate the oracle O_{tdp} in their heads:

1. At the beginning, Alice and Bob set the lists L_A, L_B to be empty.

2. Suppose it is Alice's turn to send a message μ_i. During her computation in this round, if Alice needs to know the answer of O_{tdp} to x, she reduces this query to a query to o_{tdp}. Then she uses the list L_A to run Eval and sample (or just obtain) the answer to x. Let L be the list of all query-answer pairs sampled by Eval in this round of computation till she is about to send μ_i. Alice adds L to L_A and attaches an "encryption" of the list L to μ_i (as will be described shortly). Upon receiving L, Bob will also add L to L_B before starting his computation. When it is Bob's turn to compute a message, he will do the same exact thing as described for Alice.

3. The encryption of the list L is as follows. Note that the list L has size at most $3t$ bits (because each query has t or $2t$ bits and each answer has t bits). Alice uses $3|L|$ bits from the next unused bits of the key key_2 to cover the list L and obtain L' as a one-time pad encryption of L'. Alice does not wish to reveal the length of the sequence L to whoever watches the messages being sent across (that might hurt Alice's privacy), therefore she appends enough random bits (not from the key) to make the total number of encrypted bits $3 \cdot q_2 \cdot t$ denoted by the list L''. Alice attaches L'' to her message μ_i while sending it to Bob. Since Alice wants to let Bob know how many of the blocks in the string L'' are meaningful data, she will also attach the size of the list L to μ_i in a one-time pad encrypted format using the same key key_2. The number of bits required to represent the length of L is at most $O(\log q_2) = O(\log n) < t$.

By the definition of the protocol $\overline{\Pi}$ Alice and Bob simulate the oracle o_{tdp} just as well as the lazy evaluation algorithm Eval which by Lemma 3.4 works statistically well.

Concluding Claim 3.3. In the following we show that **(1)** the protocol $\overline{\Pi}$ can indeed be broken by an adversary $\overline{\mathsf{Adv}}$ and that **(2)** the existence of $\overline{\mathsf{Adv}}$ implies the existence of Adv. These two results together prove our Claim 3.3.

Claim 3.5. *There is an adversary $\overline{\mathsf{Adv}}$ who breaks the security of $\overline{\Pi}$ without querying O_{crh}.*

Proof. In a full execution of $\overline{\Pi}$ the number of bits read from key_2 is $3tq_2$ where q_2 is the number of oracle queries made to (the simulation of) O_{tdp}. Also, in every round, there is an encrypted tag that shows the length of the list L that is sent over, and as we said this tag has length at most t. Therefore, Alice and Bob need a shared key key_2 of length $rt + 3q_2t$ for their one-time pad encryption of the lists L sent across. Alice and Bob also use a shared key key_1 of length k to begin their emulation of the protocol Π. Let $(O_{\mathsf{key}} = (\mathsf{key}_1, \mathsf{key}_2))$ be the concatenation of these keys considered as a private random oracle.

In addition to the shared keys, Alice and Bob ask q_2 queries to the oracle O_{crh} that amounts to reading q_1t random bits from this random oracle.

Therefore, Alice and Bob are reading a total of $\overline{k} = k + rt + 3q_2t + q_1t < k + 3t(r + q_1 + q_2) = k + 3t(r + q) < k + (r + q) \cdot (\log n)/\alpha$ bits from their shared random oracle $(O_{\mathsf{key}}, O_{\mathsf{crh}})$. Since we are assuming $q + r = \alpha \cdot (\frac{m-k}{\log n})$, therefore $\overline{k} < m$ and so $\overline{\mathsf{Adv}}$ exists by Theorem 1.4. □

Finally we show that breaking $\overline{\Pi}$ is at least as hard as breaking Π.

Claim 3.6. *If there is an adversary $\overline{\mathsf{Adv}}$ who breaks the security of $\overline{\Pi}$ without querying the oracle O_{crh}, then there is an adversary Adv who breaks the security of Π without querying $(O_{\mathsf{tdp}}, O_{\mathsf{crh}})$.*

Proof. Since the extra one-time pad encrypted messages attached to the usual messages μ_i are completely random, all Adv has to do over a given transcript y is to run $\overline{\mathsf{Adv}}$ over $\overline{y}$ where $\overline{y}$ is simple y augmented with $(1+3tq_2)t$ random bits attached to every μ_i in y. It is easy to see that the chance of Adv breaking Π is equal to the chance of $\overline{\mathsf{Adv}}$ breaking $\overline{\Pi}$. □

4 Lower-Bounds for Encryption and Oblivious Transfer

In this section we prove Corollary 1.3.

Corollary 1.3 (Restated)**.** *There is no fully black-box construction of semantically secure private-key encryption schemes with key length k and m bit messages using only $o(\frac{m-k}{\log n})$ oracle calls to CRHFs. For semantically secure public-key encryption schemes and oblivious transfer for m-bit messages there is no fully black-box construction using only $o(m/\log n)$ oracle calls to CRHFs.*

Proof. The corollary for private-key encryption schemes with key length k and m-bit messages follows immediately from Theorem 1.2 and the observation that k-to-m private-key amplification can be constructed with one invocation of such an encryption scheme: simply let Alice encrypt a uniformly random m-bit message X_A to Bob using the shared k-bit string as the secret key, who decrypts to obtain X_B. The completeness property of the encryption scheme implies that $\Pr[X_A = X_B] = 1 - \mathrm{negl}(n)$, and the semantic security implies that (Y, X_A) and (Y, U_m) are computationally indistinguishable, where Y is the transcript of the encryption. The corollary for public-key encryption schemes follows by an identical argument with $k = 0$.

For m-bit string oblivious transfers, the corollary follows by the fact that m-bit message *interactive* public-key encryption schemes can be constructed with one invocation of m-bit string oblivious transfers, which is proved in [GKM+00]. ([GKM+00] stated that OT implies key agreements, but their proof actually gives an interactive public-key encryption.) For the sake of completeness, we present the construction as follows: Let (S, R) be an m-bit string oblivious transfer protocol, where S takes inputs $M_0, M_1 \in \{0,1\}^m$ and R takes input bit $b \in \{0,1\}$ and learns M_b from the protocol. We define an interactive encryption scheme $(\mathrm{Sen}, \mathrm{Rec})$ where the sender Sen encrypts a message $M \in \{0,1\}^m$ to Rec by running (S, R) with S takes input $M_0 = M$ and $M_1 \leftarrow U_m$, and R takes input $b = 0$. The completeness follows by the completeness of (S, R) and we refer to [GKM+00] for the proof of security.

□

A Preliminaries

A.1 Secure Encryption

Definition A.1 (Encryption)**.** An interactive encryption scheme $(\mathrm{Sen}, \mathrm{Rec})$ is an interactive game between two randomized parties: a sender Sen and a receiver Rec who share a secret $s \xleftarrow{\$} \{0,1\}^k$. The sender receives a message $x_S \in \{0,1\}^m$ as input at the beginning of the protocol, and the receiver outputs some x_R at the end. We call the scheme $(1-\gamma)$ complete if $\Pr[x_S = x_R] \geq 1 - \gamma$ (over the randomness of the parties as well as $s \xleftarrow{\$} \{0,1\}^k$). Let y be the transcript of the interaction using x_S as sender's input. We might denote y also by $\mathrm{Enc}(x_S)$ and call it the interactive encryption of x_S. A *non-interactive* encryption scheme is an interactive encryption scheme in which the interaction consists only of a single message from Sen to Rec.

Any public-key encryption scheme is a special case of an interactive encryption scheme in which the key length k is zero.

Definition A.2 (Security of Encryption)**.** We say an adversary Adv computationally (resp. statistically) ε-breaks an (interactive or non-interactive) encryption scheme if there are two messages $x \neq x_2$ such that Adv can computationally (resp. statistically) ε-distinguish $\mathrm{Enc}(x_1)$ from $\mathrm{Enc}(x_2)$. We call an encryption scheme computationally (resp. statistically) $(1-\varepsilon)$-secure if there is no efficient (resp. computationally unbounded) adversary that ε-breaks it.

It is easy to see that any semantically secure (private-key or public-key) encryption scheme is also secure according to Definition A.2. Sine our goal is to prove *lower-bounds* we choose to work with Definition A.2.

Lemma A.3 (Sufficient Condition for Breaking Security)**.** *To ε-breaks* $(\mathrm{Sen}, \mathrm{Rec})$ *computationally (resp. statistically) it is sufficient to computationally (resp. statistically) ε-distinguish* $(X_1, \mathrm{Enc}(X_2))$ *from* $(X_1, \mathrm{Enc}(X_1))$ *where X_1 and X_2 are independently sampled random messages.*

Proof. If an adversary ε-breaks $(\mathrm{Sen}, \mathrm{Rec})$, this way by an averaging argument, there is a messages x_1 such that $\mathrm{Sen}(x_1)$ and $\mathrm{Sen}(X_2)$ (where X_2 is a random message) are ε-far. Another averaging argument shows that for some fixed x_2 it holds that $\Delta(\mathrm{Enc}(x_1), \mathrm{Enc}(x_2)) \geq \varepsilon$. □

A.2 Black-box Constructions and Separations

We employ the framework of [RTV04] but restrict ourself to type of constructions that prove "almost everywhere" security.

Definition A.4 (Black-Box Constructions)**.** A *black-box implementation* Q^P of a primitive $\mathcal{Q}$ from another primitive $\mathcal{P}$ is an oracle algorithm Q (called the implementation reduction) such that Q^P is an implementation of $\mathcal{Q}$ for any oracle P that implements $\mathcal{P}$. We say that Q^P has a *black-box security reduction*, if there exists an efficient oracle algorithm Sec such that for any oracle P implementing $\mathcal{P}$ and for any (computationally unbounded) adversary Adv who breaks the security of Q^P (as an implementation of $\mathcal{Q}$) with non-negligible advantage for some security parameter n, the oracle algorithm $\mathrm{Sec}^{P,\mathsf{Adv}}$ breaks the security of P over a polynomially related security parameter $n' = n^{\Theta(1)}$. We say that $\mathcal{Q}$ has a *fully black-box construction* from $\mathcal{P}$ if there is a black-box implementation Q and a black-box proof of security Sec as above.

Definition A.5 (Security Threshold)**.** A primitive $\mathcal{P}$ has *security threshold* $\tau_{\mathcal{P}}$ if an adversary "breaking" $\mathcal{P}$ has to "win" in the security game of $\mathcal{P}$ with probability $\tau_{\mathcal{P}} + \varepsilon$ for a non-negligible ε.

For simplicity, in this work we restrict ourself to constructions that preserve the security parameter (after a possible re-scaling of it), in which case we would get $n' = n$ in Definition A.4.

Definition A.6. Let $\mathcal{O}$ be a *set* of *randomized* oracles (e.g., the set of all partially-fixed random oracles). We say that a primitive $\mathcal{P}$ (e.g., one-way functions or CRHFs) has a secure black-box implementation using $\mathcal{O}$, if there is a $\mathrm{poly}(n)$-time implementation algorithm $P^{\mathbf{O}}$ for $\mathcal{P}$ such that for every randomized oracle $\mathbf{O} \in \mathcal{O}$, any computationally-unbounded adversary A who (only) knows the distribution of $\mathbf{O}$ and asks $\mathrm{poly}(n)$ queries to $\mathbf{O}$ can break $P^{\mathbf{O}}$ with advantage at most $\mathrm{negl}(n)$ (above the security threshold $\tau_{\mathcal{P}}$).

A similar argument to that of Lemma A.7 below is implicit in [BMG07] and [DLMM11], but the form we use below is due to [MP12].

Lemma A.7. *Let $\mathcal{P}$ and $\mathcal{Q}$ be two cryptographic primitives and $\tau_{\mathcal{P}} = 0$. For a randomized oracle $\mathbf{O}$, suppose one can break the black-box security of* any *implementation $Q^{\mathbf{O}}$ of $\mathcal{Q}$ with non-negligible probability by asking* $\mathrm{poly}(n)$ *oracle queries to $\mathbf{O}$. Suppose also that there exists a black-box secure implementation P of $\mathcal{P}$ from $\mathbf{O}$. Then there is no fully black-box construction of $\mathcal{Q}$ from $\mathcal{P}$.*

A.3 Shannon's Lower-Bound

In his seminal work, Shannon [Sha49] showed that in any *perfectly secure* encryption scheme, the key space should be as large as the message space. Shannon's lower-bound can be extended to obtain an actual (statistical) attack if the key is even one bit shorter than the encrypted message (see [Pas] for a proof).

Theorem A.8. *For any $(1-\varepsilon)$ statistically secure encryption scheme with completeness $1-\gamma$ for key space $\mathcal{K}$ and message space $\mathcal{M}$ it holds that $|\mathcal{K}| \geq (1-\varepsilon-\gamma)|\mathcal{M}|$.*

Note that when $\varepsilon+\gamma < 1/2$ Theorem A.8 implies that $k \geq m$ where $\mathcal{K} = \{0,1\}^k$ and $\mathcal{M} = \{0,1\}^m$. For completeness we present a proof by adopting the proof of [Dod12] to also handle randomized decryption.

Proof. For sake of contradiction, we assume that $\Delta((X_1, \mathrm{Enc}(X_2)), (X_1, \mathrm{Enc}(X_1)) \leq \varepsilon$ where X_1 and X_2 are independently chosen random messages.

Let Eve be an algorithm that receives (x, y), then for all $S \in \mathcal{K}$ it runs $\mathrm{Rec}(y)$ with independent decryption randomness, and outputs 1 if it obtains x in any of its $|\mathcal{K}|$ executions. Let $\alpha = \Pr_{S,X_1,X_2}[\mathrm{Eve}(X_1, \mathrm{Enc}^S(X_2)) = 1]$. Since $\Delta((X_1, \mathrm{Enc}(X_2)), (X_1, \mathrm{Enc}(X_1)) \leq \varepsilon$, thus $\alpha \geq \Pr_{S,X_1}[\mathrm{Eve}(X_1, \mathrm{Enc}^S(X_1)) = 1] - \varepsilon$. But since Eve will always try the actual key S, by $(1-\gamma)$ completeness it holds that $\Pr_{S_1,X_1}[\mathrm{Eve}(X_1, \mathrm{Enc}^{S_1}(X_1)) = 1] \geq 1-\gamma$, and so $\alpha \geq 1-\varepsilon-\gamma$.

On the other hand, for every fixed key $s \leftarrow S$ that Eve uses, over the randomness of X_2 Eve outputs 1 only with probability at most $1/|\mathcal{M}|$ (because it does not have any information about X_2). Therefore, by a union bound it holds that $\alpha \leq |\mathcal{K}|/|\mathcal{M}|$, and so $|\mathcal{K}|/|\mathcal{M}| \geq 1-\varepsilon-\gamma$. □

B Two-Sided Lower-Bounds for Non-Interactive Encryption

The following theorem improves our Theorem 1.4 for the special case of non-interactive k-to-m key-amplification schemes by proving separate lower-bounds on the number of oracle bits read Alice and Bob (rather than their unions) Note that this theorem still generalizes Shannon's result (Theorem A.8) to the case of private random oracles.

First we formalize the notion of non-interactive encryption.

Definition B.1 (Encryption Using Random Oracles)**.** In an (interactive or non-interactive) encryption scheme using a random oracle both Sen and Rec access a random oracle O; $\mathrm{Sen}^O(x_S)$ reads k_S bits of O, and Rec^O reads k_R bits of O to finally get x_R. The completeness is defined similarly to Definition A.1. The security is defined similarly to Definition A.2 while we do *not* allow the adversary to access the random oracle O. We call k_S and k_R, in order the *(bit) query complexity* of the sender and the receiver. We also call the number of bits of O read by either of Alice or Bob the key length and denote it by k.

It is easy to see that a k-to-m private-key amplification scheme with a single message sent from Alice to Bob is in fact a non-interactive encryption scheme as defined above and vice versa.

Theorem B.2. *For any $(1-\varepsilon)$-secure non-interactive encryption scheme with completeness $1-\gamma$ and message space $\mathcal{M} = \{0,1\}^m$ using a private random oracle the following two hold*

1. $2^{k_R} \geq 2^m \cdot (1-\varepsilon-\gamma)$
2. $2^{k_S} \geq 2^m \cdot (1-\varepsilon-\gamma)$

where k_S (resp. k_R) is the number of bits of the oracle read by the sender (resp. receiver).

Proof. For sake of contradiction assume that $\Delta((X_S, \mathrm{Enc}(X'_S)), (X_S, \mathrm{Enc}(X_S))) \leq \varepsilon$ where X'_S is a random message chosen indecently of random message X_S. We first prove $2^{k_R} \geq 2^m \cdot (1-\varepsilon-\gamma)$ using a hybrid argument.

- **Experiment** Real**.** In this game the sender chooses $x_S \leftarrow \{0,1\}^m$ at random, executes $\mathrm{Sen}^O(x_S)$ using a random oracle O, and then it sends $y = \mathrm{Sen}^O(x_S)$ to the receiver who executes $\mathrm{Rec}^O(y)$ to obtain x_R. In this game it holds that $\Pr_{\mathsf{Real}}[x_S = x_R] = 1 - \gamma$ by the $(1-\gamma)$-completeness of the scheme.

- **Experiment** Hyb_1**.** In this game, the sender again samples $x_S \xleftarrow{\$} \mathcal{M}$ at random, but then it samples $y = \mathrm{Enc}(x_S)$ from the same distribution as in Real conditioned on x_S *directly* without sampling O. Then it samples O conditioned on x_S, y. The receiver gets access to the sampled O and obtains $x_R = \mathrm{Rec}^O(y)$. Since we just sampled things in a different order, we still have $\Pr_{\mathsf{Hyb}_1}[x_S = x_R] \geq 1 - \gamma$.

- **Experiment** Hyb_2**.** This experiment is the same as Hyb_1 with the difference that y is sampled from the distribution $\mathrm{Enc}(x'_S)$ for an independently sampled message x'_S. We still sample O conditioned on x_S, y from the same conditional distribution as in Hyb_1. Since we assumed $\Delta((X_S, \mathrm{Enc}(X'_S)), (X_S, \mathrm{Enc}(X_S)))$ (for random independent variables X_S and X'_S) in this experiment we obtain $\Pr_{\mathsf{Hyb}_2}[x_S = x_R] \geq 1 - \gamma - \varepsilon$.

- **Experiment** Indep**.** In this experiment the receiver does not access O, and instead it tosses k_R coins and uses them as the oracle answers. Note that for every execution of Hyb_2 that leads to $x_S = x_R$, here the same view of the receiver will occur (matching the answers of O) with probability 2^{-k_R}, and so it holds that $\Pr_{\mathsf{Indep}}[x_S = x_R] \geq (1 - \gamma - \varepsilon)2^{-k_R}$. However, in this experiment the message Y (as a random variable) is independent of X_S and the receiver does not access O, so the probability that $x_S = x_R$ is at most 2^{-m} (due to the uniform choice of x_S). Thus it holds that $(1 - \gamma - \varepsilon)2^{-k_R} \leq 2^{-m}$ which finishes the proof of the first part of Theorem B.2.

The proof of the second part $2^{k_S} \geq 2^m \cdot (1 - \varepsilon - \gamma)$ goes through a different hybrid argument.

- **Experiment** Real**.** The same as the experiment Real above.

- **Experiment** Hyb_1**.** In this game, the sender again samples $x_S \xleftarrow{\$} \mathcal{M}$ and $y = \mathrm{Enc}(x_S)$ *directly* without sampling O as in the previous Hyb_1 above. Then we directly sample O_S to be a sequence of k_S bits that the sender encounters in its execution of $\mathrm{Sen}^O(x_S)$ *conditioned* on obtaining y. Then we sample $\overline{O}_S$ as a sequence of independent random bits to complement O_S to get a full oracle O. Then the receiver will use O to execute $\mathrm{Rec}^O(y)$ to obtain x_R. Since we just sampled things the same was as in Real (in a different order, but from their conditional marginal distribution) we still have $\Pr_{\mathsf{Hyb}_1}[x_S = x_R] \geq 1 - \gamma$.

- **Experiment** Hyb_2**.** This experiment is a modification of Hyb_1 exactly the same way that Hyb_2 modified Hyb_1 in the above case. Namely, Hyb_2 is the same as Hyb_1 with the difference that y is sampled from the distribution $\mathrm{Enc}(x'_S)$ for an independently sampled message x'_S. Note that we still sample O_S conditioned on x_S, y using the same conditional distribution as in Hyb_1. The partial oracle $\overline{O}_S$ is again chosen independently of the previously sampled components: (x_S, y, R_S) to complement O_S. Since we assumed $\Delta((X_S, \mathrm{Enc}(x'_S)), (X_S, \mathrm{Enc}(X_S)))$ in this experiment we again obtain $\Pr_{\mathsf{Hyb}_2}[x_S = x_R] \geq 1 - \gamma - \varepsilon$.

- **Experiment** Indep**.** This experiment is the same as Hyb_2 with the difference that here the sender does not use the sampled oracle O_S in its full oracle O sent to the receiver: it tosses k_S coins, denoted as partial oracle O'_S, and uses them instead of O_S. (The partial oracle $\overline{O}_S$ is also sampled uniformly at random just as before). We note that for every execution of Hyb_2 that leads to $x_S = x_R$, here the same view of the receiver will occur (due to $O_S = O'_S$) with

probability at least 2^{-k_R}, and so it holds that $\Pr_{\mathsf{Indep}}[x_S = x_R] \geq (1-\gamma-\varepsilon)2^{-k_S}$. However, in this game the message y that the receiver receives is indecent of x_S and it also uses a random oracle $O = (O'_S, \overline{O}_S)$ which is independent of (x_S, y), so the probability that $x_S = x_R$ can be at most 2^{-m} (due to the uniform choice of x_S). Therefore it holds that $(1-\gamma-\varepsilon)2^{-k_S} \leq 2^{-m}$ which finishes the proof of the second part of Theorem B.2.

□

C Omitted Proofs

Proof of Lemma 2.8. Let $P_{XY}(\cdot,\cdot)$ denote the pmf of (X,Y), and $P_Z(\cdot)$ be the pmf of Z. Let $P_Y(y) \triangleq \sum_x P(x,y)$ be the marginal pmf of Y. We modify the distribution (X,Y) by the following iterative procedure.

- Let $S_+ = \{y \in [M] : P_Y(y) > P_Z(y)\}$ and $S_- = \{y \in [M] : P_Y(y) < P_Z(y)\}$. Repeat the following until Y has the same distribution as Z.
  - Selects $y_+ \in S_+$, $y_- \in S_-$, and $x \in [N]$ such that $P_{XY}(x, y_+) > 0$.
  - Let $\delta = \min\{P_{XY}(x,y_+), (P_Y(y_+) - P_Z(y_+)), (P_Z(y_-) - P_Y(y_-))\}$, and modify (X,Y) by $P_{XY}(x,y_+) := P_{XY}(x,y_+) - \delta$ and $P_{XY}(x,y_-) := P_{XY}(x,y_-) + \delta$.
  - Update the sets S_+ and S_- accordingly.

Note that the above procedure does not change the marginal distribution of X. Thus, by definition, if the above procedure terminates, we obtain a desired distribution (X,Z).

Also note that the probability mass of each $y \in S_+$ (resp., $y \in S_-$) is monotonically decreased (resp., increased) towards $P_Z(y)$, and once $P_Y(y) = P_Z(y)$, y is removed from S_+ (resp., S_-) and the probability mass of $P_{XY}(x,y)$ for all $x \in [N]$ will not be modified throughout the procedure. Furthermore, at the end of each iteration, either $P_{XY}(x,y_+)$ becomes 0, or the size of S_+ or S_- is reduced by 1. Since there are only finitely many (x,y_+) and the size of S_+ and S_- are finite, the procedure terminates in some finite number t of iterations.

Let $(X,Y_0) = (X,Y)$ and $(X,Y_1),(X,Y_2),\ldots,(X,Y_t)$ denotes the modified distribution at the end of each iteration. We have $(X,Y_t) = (X,Z)$. It is not hard to verify that (1) by the choice of δ, we have $\Delta((X,Y_i),(X,Y_{i+1})) = \Delta(Y_i,Y_{i+1})$ for every $i \geq 0$, and (2) $\Delta(Y,Z) = \sum_{i=0}^{t-1}\Delta(Y_i,Y_{1+1})$ since the probability mass of every $y \in [M]$ is monotonically increased or decreased towards $P_Z(y)$ throughout the procedure. It follows that

$$\Delta(Y,Z) \leq \Delta((X,Y),(X,Z)) \leq \sum_{i=0}^{t-1}\Delta((X,Y_i),(X,Y_{i+1})) = \sum_{i=0}^{t-1}\Delta(Y_i,Y_{i+1}) = \Delta(Y,Z).$$

□

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